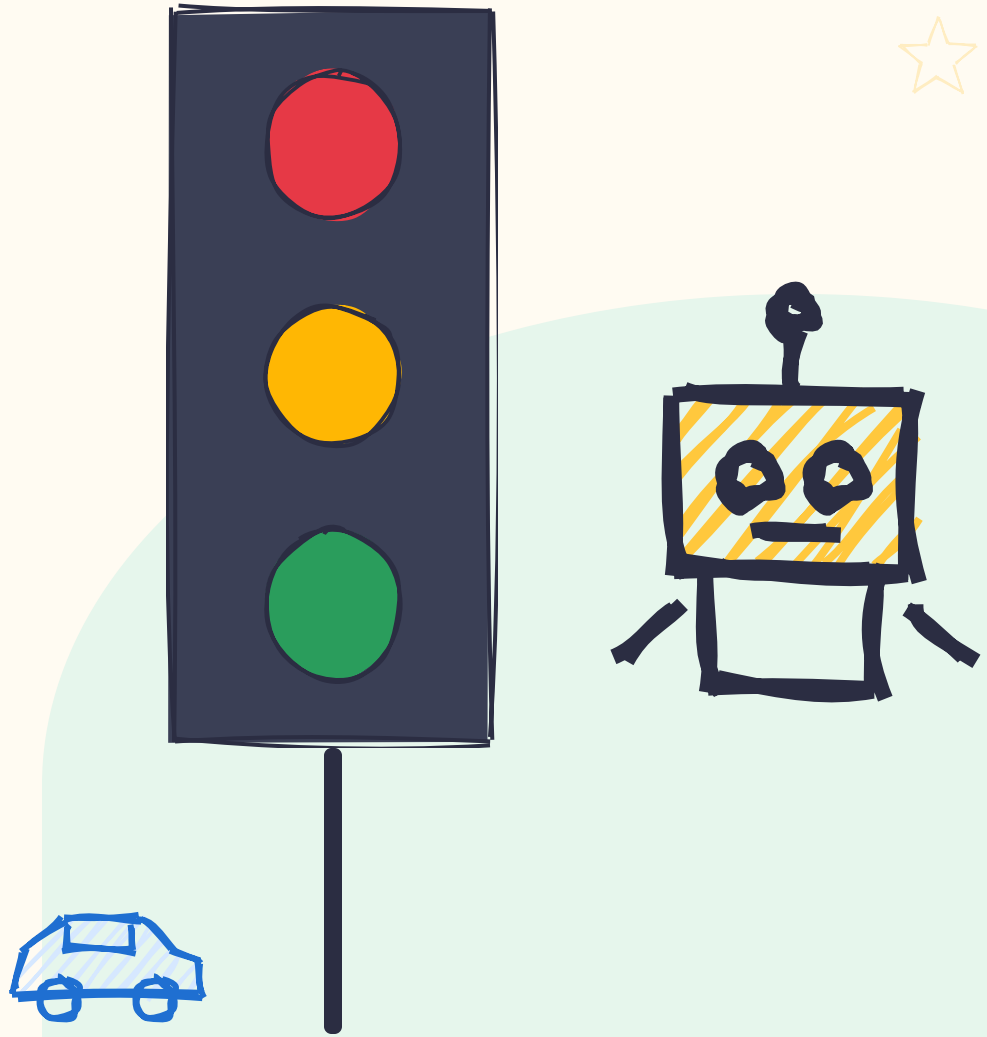
Little Explorers · Week 4

Traffic Light Controller

Let's help cars and people stay safe!

Ages 5-7 | Arduino + picture blocks | 60 minutes



Did you know?
The first electric traffic light was switched on in 1914, more than 100 years ago!

Let's think...

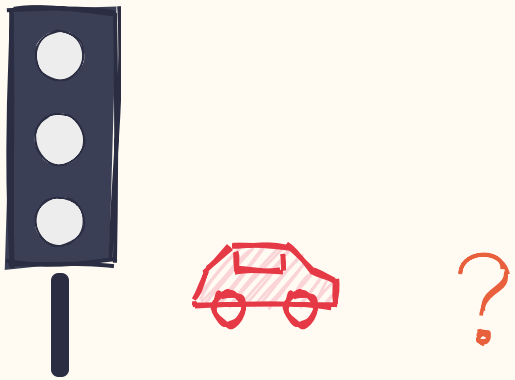
Uh-oh! The traffic lights stopped working!

Cars are coming from every side.
What will happen now?

Who goes first? **Can I cross safely?** **Will cars crash?**

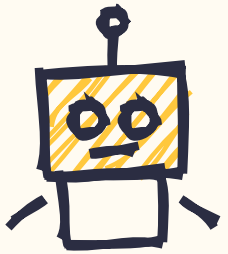


In real life
When lights break, a traffic police officer stands in the road and tells cars when to stop and go.



Our learning aim

Today, I will learn to...



1 I can tell what **red**, **yellow** and **green** mean.

2 I can put the lights in the **right order**.

3 I can make a real **robot brain (Arduino)** run my traffic light.

In real life

Engineers program real traffic lights in every city so millions of people stay safe every day.

Watch and think

Video time!



How Do Traffic Lights Work?

STEMRocket · about 2 minutes · youtu.be/SWJP-RXOk6M

Watch for...

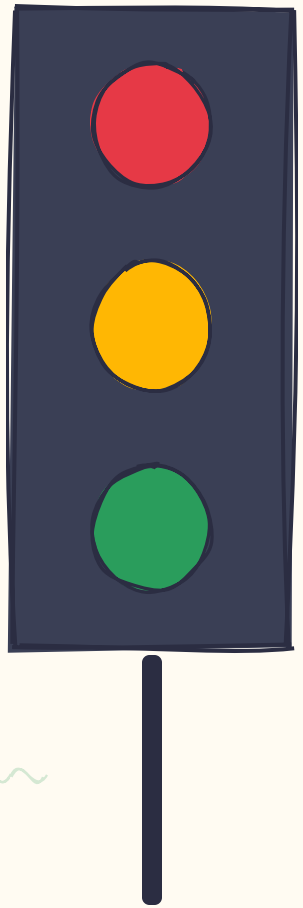
1. What does **red** mean?
2. What does **green** mean?
3. Who tells the lights when to change?

Scan to watch

After the video: Play 'traffic lights' with your partner: red = freeze, amber = get ready, green = walk!

Look closely

Meet the traffic light



RED
STOP!

YELLOW
Get ready / slow down

GREEN
GO!



In real life
Look for traffic lights on your way home today. Which colour did you see?

Not just for cars

Traffic lights help people too!



Red person

Stand and wait



Green person

Now you can walk





Safety rule

Always cross on the green person and hold a grown-up's hand.

Look around you

Lights that talk to us are everywhere!

Railway crossings

Red lights flash when a train is coming

Airports

Runway lights guide planes to land safely

Chargers at home

Red = charging, green = full







Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Red Light, Green Light game
2. Meet our tools
3. Let's code our traffic light
4. Can you do it?

Stand up and play!

Red Light, Green Light game

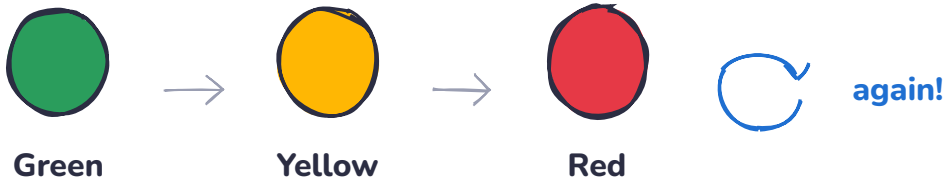
GREEN: walk on the spot

YELLOW: walk very slowly

RED: freeze like a statue!

Your teacher shows the colours **in order**. Can you move the right way every time?

The order is called a sequence



In real life

Real traffic lights repeat the same order all day and all night. Repeating is called a loop.

Our builder's kit

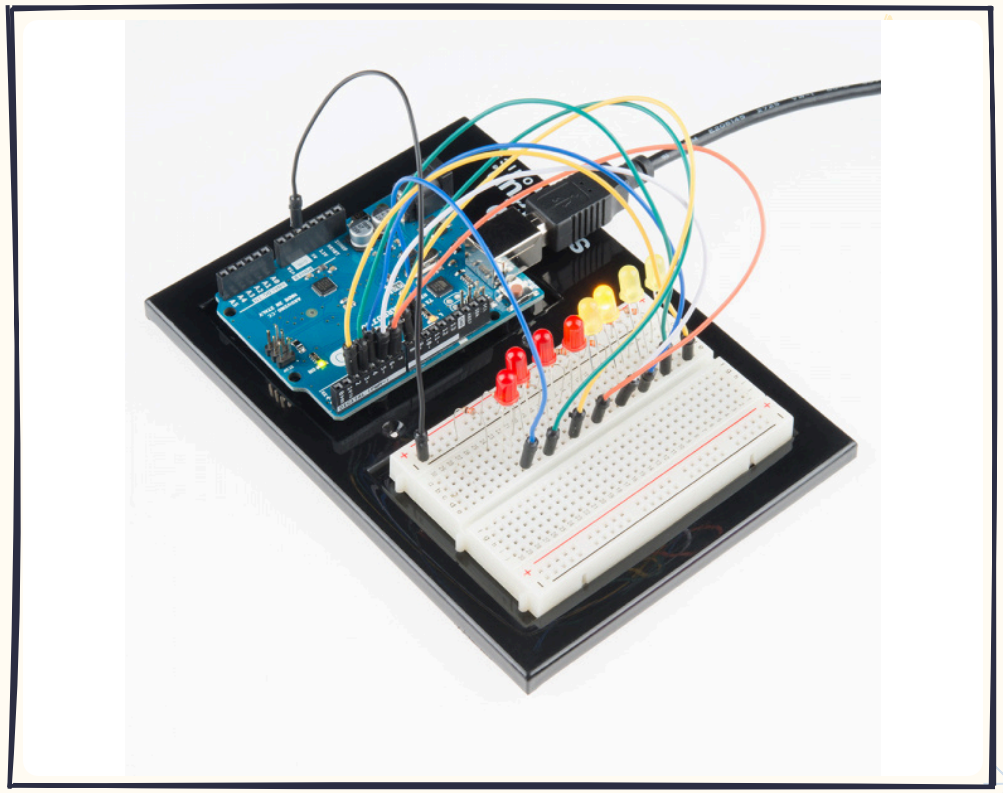
Meet our tools

Arduino
The robot brain that follows our instructions

3 LEDs
Tiny lights: red, yellow and green

Wires
Roads for electricity to travel on

Laptop
Where we snap our picture blocks



Teacher: the circuit is already wired. Children only plug in the USB cable and program.

In real life
A tiny computer like our Arduino is also inside real traffic lights, washing machines and toys!

MAKE

Time to build!

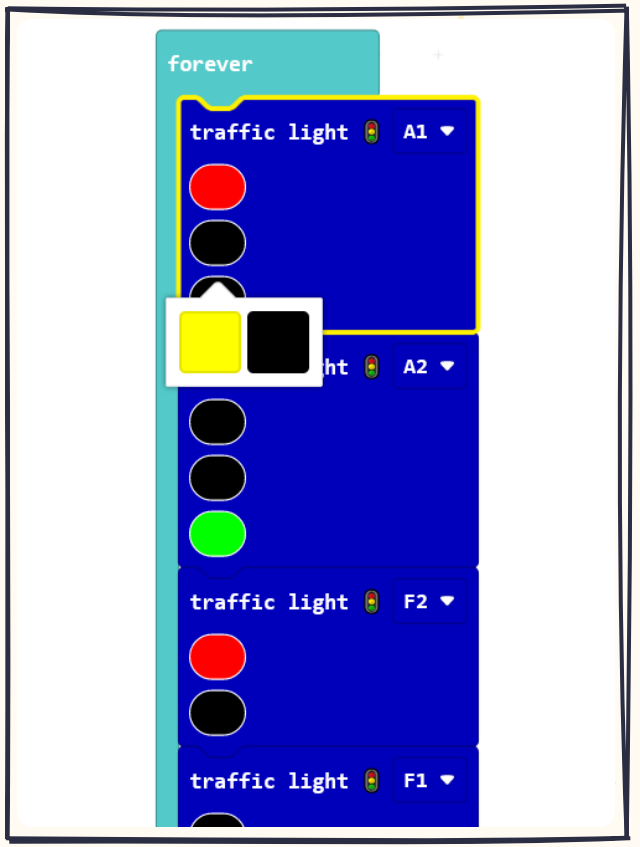
Let's code our traffic light

```
when Arduino starts
forever
    turn ON green light
    wait 3 seconds
    turn OFF green light
    turn ON yellow light
    wait 1 second
    turn OFF yellow light
    turn ON red light
    wait 3 seconds
    turn OFF red light
```

3 easy steps

1. Snap the blocks **in order**
2. Press **Upload**
3. Watch your lights!

Magic word: WAIT
The wait block tells each light how many seconds to stay on.



Blocks look like this in our coding tool

Challenge time!

Can you do it?



Safe crossing: make green longer so children have time to cross.



Beep beep: add a buzzer that beeps when it is safe to walk.



Bug hunt: the lights are in the wrong order. Can you fix them?

What did we learn?

- ☑ Red means stop, yellow means get ready, green means go
- ☑ Instructions must be in the right order (a sequence)
- ☑ A robot brain follows our blocks exactly

Tell your family tonight

Teach them what each traffic light colour means, and show them the green person when you cross together!



Next week: an Automatic Night Lamp that switches on in the dark!